

Herbal micelles-loaded ROS-responsive hydrogel with immunomodulation and microenvironment reconstruction for diabetic wound healing

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ABSTRACT

Persistent inflammation is a major cause of diabetic wounds that are difficult to heal. This is manifested in diabetic wounds with excessive reactive oxygen clusters (ROS), advanced glycation end products (AGE) and other inflammatory factors, and difficulty in polarizing macrophages toward inhibiting inflammation. Berberine is a natural plant molecule that inhibits inflammation; however, its low solubility limits its biological function through cytosol. In this study, we designed F127 micelles to encapsulate berberine with the aim of improving its solubility and bioavailability. Meanwhile, in order to achieve effective drug delivery at the wound site, we designed an injectable ferrocene-cyclodextrin self-assembled oxidation-reactive supramolecular hydrogel drug delivery system. Cellular experiments have shown that the hydrogel can reduce intracellular ROS and AGE production, attenuate cellular damage, promote macrophage polarization toward inhibition of inflammation, and reduce the secretion of inflammatory factors. In an animal model of diabetic mice, this hydrogel dressing reduces the level of inflammation in diabetic wounds, optimizes collagen deposition in diabetic wounds, and ultimately achieves high-quality diabetic wound healing. The work offers a straightforward and effective solution to the challenge of administering hydrophobic anti-inflammatory agents in the context of diabetic wound therapy.

1. Introduction

Wound healing in the skin consists of four phases hemostasis, inflammation, proliferation, and remodeling, each of which occurs in a sequential and effective manner with some spatial and temporal overlap to ensure efficient healing of skin wounds [1–3]. When the various phases of wound healing do not proceed properly, chronic wounds such as diabetic wounds can result [4]. The main cause of diabetic wounds that are difficult to heal is an imbalance in the inflammatory phase of wound healing due to excess glucose in the wound environment, i.e., high levels of advanced glycation end-products (AGE) resulting in an excess of reactive oxygen clusters (ROS) and inflammatory factors, and the inability of macrophages to switch from pro-inflammatory M1 to anti-inflammatory M2, resulting in an inability to transition smoothly

from the pro-inflammatory phase to the next phase of the wound healing process and ultimately prolonging wound healing [5–8]. Therefore, there is a need to develop a wound dressing that regulates the inflammatory phase of wound healing for better treatment of chronic diabetic wounds [9].

In recent years, natural derivatives, especially those of plant origin, have been widely used in the treatment of chronic wounds due to their efficacy, low cost and easy availability [10]. Berberine (Ber) is a herbal component of the Chinese medicine *Berberis vulgaris*. Berberine has been shown to have excellent anti-inflammatory properties, antimicrobial properties, and the ability to promote wound healing, and has also been used as a prescription drug in China [11–13]. The anti-inflammatory ability of berberine works mainly by inhibiting AGE and modulating ROS production, reducing the levels of

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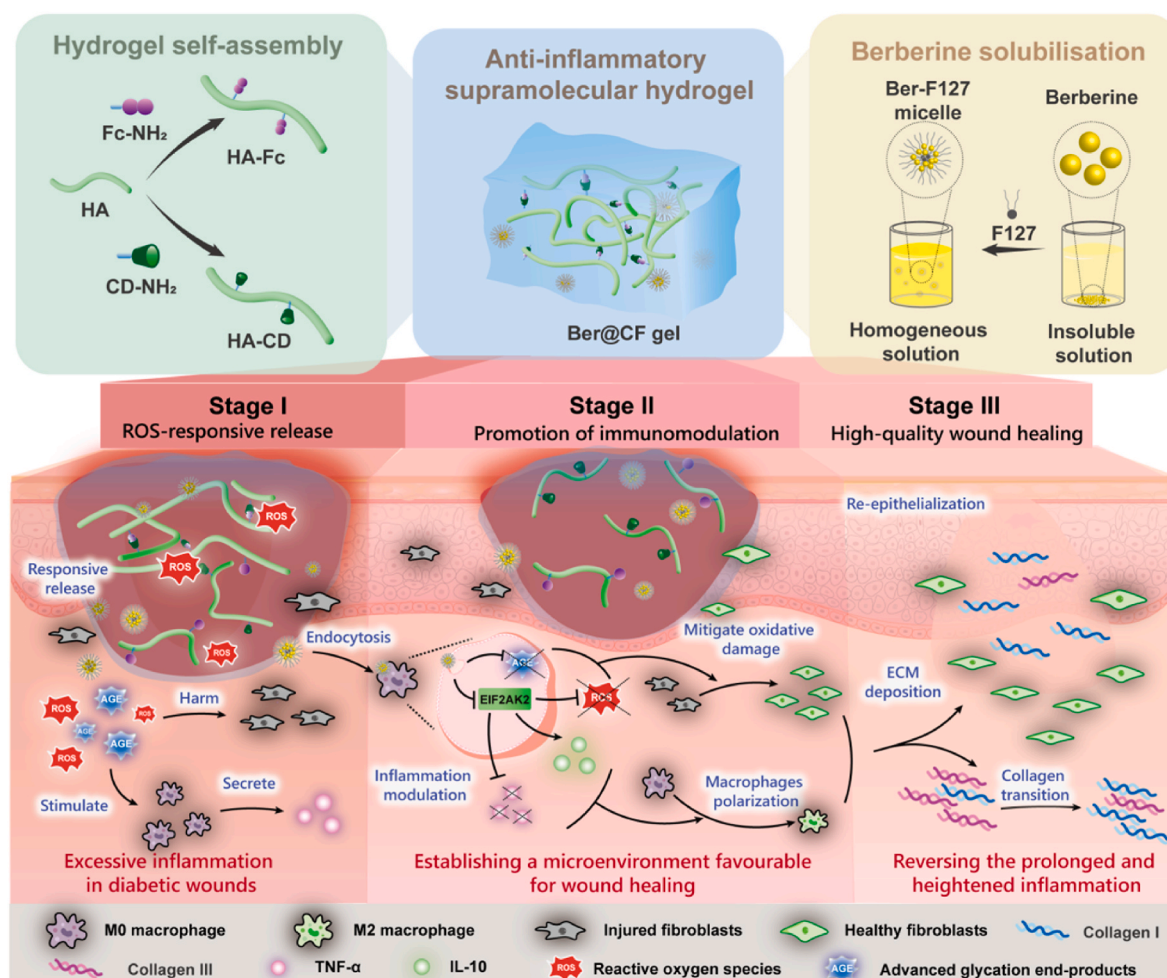
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pro-inflammatory factors such as TNF- α and increasing the levels of anti-inflammation factors such as IL-10, while promoting the shift of macrophages toward inhibiting inflammation [14–16]. However, the poor water solubility of berberine limits its efficacy by direct administration and there is a need for a better way to deliver the drug [17]. There have been several reports on improving the solubility of difficult drugs such as berberine, such as self-assembled nanogels [18] and microgels [19]. In addition, micelles are amphiphilic polymers that self-assemble into microencapsulated structures, and the water solubility of hydrophobic drugs can also be improved by micellar encapsulation. Meanwhile, compared with other solutions, after the drugs are encapsulated by the micelles, the cells can better take up the drugs through endocytosis, so as to improve the therapeutic effect of the drugs [20,21]. The commercialized Pluronic F127 micelles were chosen for the construction of berberine micelles due to their good biocompatibility and low critical micelle concentration [22,23].

Hydrogels are cross-linked hydrophilic polymers with a three-dimensional reticulated porous structure [24]. Hydrogels have been employed as wound dressings due to their exceptional moisturizing, exudate absorption, wound protection, and drug delivery capabilities [25–27]. In recent years, a variety of hydrogel formulations have been developed based on the physiological microenvironment of diabetic wounds with the objective of facilitating the responsive release of drugs for the treatment of diabetic wounds [28–30]. Based on the complex physiological microenvironment of diabetes, a variety of responsive-release drugs such as pH-responsive, ROS-responsive, and glucose-responsive hydrogels have been designed for diabetic wound

therapy [31–34]. By responsive release of drugs, on-demand release of drugs at the wound and thus efficient utilization can be achieved [35]. In addition, the hydrogels should have good injectability and self-healing properties to fulfill the need of filling irregular wounds [36–38]. The construction of the aforementioned hydrogels is currently achieved through a number of established techniques, including host-guest self-assembly, phenyl borate bonding, Schiff base bonding, ion chelation, and others [39]. Among these, the use of cyclodextrin-based host-guest self-assembled hydrogels has been the subject of considerable research and has gained widespread application [40,41].

Based on the above, we designed a hydrogel dressing for chronic wound treatment as shown in Scheme 1. Sodium hyaluronate was utilized as the fundamental structure for the grafting of a ferrocene group (Fc), which exhibits oxidative reactivity, and a cyclodextrin group (CD), capable of self-assembling with the ferrocene group, respectively. The straightforward amalgamation of these two solutions yielded a supramolecular hydrogel [42–44]. The supramolecular hydrogel dressing was disassembled by ROS attack in the physiological microenvironment of chronic wounds, thus oxidatively responding to the release of the anti-inflammatory drug berberine and realizing the on-demand treatment of chronic wounds [45]. To solve the problems of poor water solubility and low therapeutic efficacy of berberine, we prepared the amphiphilic polymer F127 into micelles wrapped with hydrophobic anti-inflammatory agent berberine by the hydrated film method to improve the water solubility and therapeutic efficacy of berberine. The ferrocene-cyclodextrin-hyaluronic acid hydrogel (CF gel) oxidative-responsive delivery of berberine-F127 micelles (Ber-F127)



Scheme 1. Hyaluronate-based ferrocene-cyclodextrin hydrogel encapsulating Ber-F127 drug micelles for chronic wound treatment.

was targeted to the treatment of chronic wounds, which reduced the content of AGE, ROS and inflammatory factors at the chronic wounds, promoted the shift of macrophages to the M2 type, facilitated the transition from the inflammatory phase of wounds to the proliferative and remodeling phases, and facilitated the healing of wounds.

2. Materials and methods

2.1. Materials

Hyaluronic acid (HA, Mw = 20–40 w) was purchased from Bloomage Biotechnology Co.Ltd (China). β -Cyclodextrin (β -CD), 1,6-Hexanedi-amine, N,N'-Carbonyldiimidazole(CDI), 4-(4,6-Dimethoxy-1,3,5-triazin-2-yl)-4-methyl morpholinium chloride (DMTMM), 2-Morpholinoethanesulfonic acid hydrate (MES), Triethylamine (TEA), benzo triazole-1-yl- oxytripyrrolidinophosphonium hexafluorophosphate (PyB op, 98 %), Trifl Tetraethylammonium bromide (TBA), 1,1-diphenyl-2-picrylhydrazyl (DPPH), Ferrocenyl methylamine (98 %), berberine hydrochloride were purchased from Energy Chemical (China). Pluronic® F-127 (F127), 2,7-Dichlorodihydrofluorescein diacetate (DCFH-DA) and Streptozocin (STZ) were purchased from Sigma-Aldrich LLC (USA). Methylthiazolyl-diphenyl-tetrazolium bromide (MTT), LIVE/DEAD® Viability/Cytotoxicity Kits, Fluorescein Phalloidin, TNF- α (Tumor Necrosis Factor- α) Elisa Kits were purchased from Solarbio Life Sciences (China). IL-10 (Interleukin-10) Elisa Kits, RAGE Elisa Kits were purchased from Shanghai ZCIBIO Technology Co., Ltd (China). Rabbit Anti-EIF2AK2 antibody, Goat Anti-Rabbit IgG H&L(Cy3 conjugated) were purchased from Beijing Bioss (China). L929 cells and RAW264.7 cells were obtained from American Type Culture Collection (ATCC, Rockville, MD). RPMI-1640 cell medium, DMEM cell medium, fetal bovine serum (FBS), and penicillin-streptomycin (P/S) were purchased from Thermo Fisher Scientific Inc. Male C57BL/6J mice (25–30 g) were purchased from Beijing Vital River Laboratory Animal Technology Co., Ltd. (China). Isoflurane was purchased from Beijing Ruitaibio Technology Co., Ltd (China).

2.2. Synthesis of Ber-F127 micelles

With reference to the literature [46], Ber-F127 micelles were successfully synthesized by the hydrated film method as follows. 16 mg of berberine hydrochloride was dissolved in chloroform 6 mL of and 120 mg of F127 was dissolved in 6 mL of methanol, which were mixed in the ratio of 1:1 (v/v) and stirred for 2 h to allow full interaction of berberine and F127. Subsequently, the mixed solution was rotary evaporated under vacuum at 35 °C to obtain a drug-loaded yellow polymer film. The obtained product was dried in a vacuum drying oven for more than 12 h. After ensuring that the polymer film was completely dry, the film was rehydrated in deionized water at 40 °C with continuous stirring to assemble the micelles. After homogeneous dispersion, the micelle suspension was centrifuged at 3500 rpm for 10 min to initially separate the uncoated berberine, and the solution of berberine micelles was further filtered through a 0.45 μ m membrane to ensure a uniform distribution of micelle particle size. Finally, Ber-F127 micelles were freeze-dried to obtain an orange-yellow powder, and stored at 4 °C under light protection.

2.3. Synthesis of CF gel and Ber@CF gel

HA-CD and HA-Fc were synthesized according to references [47], and the obtained products were dissolved in deuterium aqueous solution (D_2O), and the synthesis of both was verified by 400 MHz nuclear magnetic resonance (1H NMR). The synthesized HA-CD and HA-Fc were dissolved in PBS buffer with 5.5 % and 4.5 % solution, respectively, and the CF gel was successfully obtained by co-mixing in the ratio of 1:1 (v/v) for half an hour. The Ber@CF gel was obtained by dissolving different concentrations of berberine in the precursor PBS buffer of

HA-CD and HA-Fc, and then mixing them according to the above-mentioned steps. After that, it was mixed according to the above steps.

2.4. Characterization of Ber-F127 micelles

Free berberine and Ber-F127 micelles were dissolved in water at a concentration of 2 mg/mL of berberine, and the dissolution conditions were photographed. Berberine fluoresces at an excitation wavelength of 460 nm and an emission wavelength of 390 nm, and the above solution was photographed with a Thunder microscope (Leica, Germany) to observe its precipitation and dispersion.

The morphology of micelles was observed by transmission electron microscope (TEM, HT7800, Hitachi) and the particle diameter was measured by dynamic laser light scattering (DLS, Zetasizer Nano, Malvern Panalytical, UK).

The amount of Berberine in micelles was further measured. Berberine was dissolved in DMSO with a series concentration and their absorbance at 350 nm was measured by the microplate reader (Cytation 3, BioTeK, USA). to obtain the standard curve. The concentration of Ber-F127 dissolved in DMSO was quantified through the standard curve. The drug encapsulation efficiency and the drug loading efficiency were calculated according to the following equation:

$$\text{Drug encapsulation (\%)} = W_{\text{berberine}}/W_{\text{initial}} \times 100\%$$

$$\text{Drug loading (\%)} = W_{\text{berberine}}/W_{\text{Ber-F127}} \times 100\%$$

Where $W_{\text{berberine}}$ was the weight of berberine in Ber-F127, W_{initial} was the weight of berberine fed initially and $W_{\text{Ber-F127}}$ was the weight of Ber-F127.

Endocytosis of micelles was verified with L929 cells. L929 cells were inoculated at a concentration of 1×10^5 /mL and cultured in confocal dishes for 24 h to ensure complete apposition, and free berberine and Ber-F27 micelles were co-cultured again with L929 cells at a concentration of 20 μ g/mL for 24 h. Afterwards, L929 cells were labeled with Dil dye to label the cell membrane. The intensity of berberine fluorescence in the cell membrane and the position of the cell membrane were observed with a Thunder microscope to determine the endocytosis of the drug.

The effect of micelles on cellular efficacy was judged by micelles on EIF2AK2 water of RAW264.7 cells after LPS stimulation. The procedure is the same as the corresponding section 2.11.

Improvement of berberine efficacy after micellization explored by a diabetic mouse model. The male C57BL/6J mice were induced to diabetes by STZ. Briefly, after fasting for 12 h, the mice were received intraperitoneal injections with STZ in citrate buffer solution (0.1 M, pH 4.5) at a dose of 160 mg/kg body weight. The blood glucose and weight of each mouse were measured, and all mice exhibited severe hyperglycemia (>16.7 mM) for a week. 4 mice were randomly divided into two groups (2 mice per group): free berberine and Ber-F127 micelle. Berberine was administered at a concentration of 2 mg/mL. Berberine concentrations were the same in subsequent animal experiments. The mice were anesthetized with isoflurane and the hair on their backs was shaved. Two 8 mm circular full-thickness wounds were made on each side of the back and then 50 μ L prepared materials were respectively applied to the wounds after the surgery. Ultimately, the wound surface was covered with the 3 M Tegaderm TM film. In vivo fluorescence images were obtained using an *in vivo* imaging system (Caliper Life Sciences, USA) by DCFH-DA labeling on Day 0, 2 and 4.

2.5. Physical characterizations of hydrogels

The dispersion status of Ber-F127 micelles in hydrogels was obtained by observing the fluorescence intensity and dispersion status of berberine in hydrogels by Thunder microscope.

The hydrogels were freeze-dried and placed in liquid nitrogen quenched and cooled for 60 s and then broken, and the cross sections

were used to observe the microstructure of the hydrogels. The microstructures of lyophilized CF gel, Ber@CF gel were observed by the scanning electron microscope (SEM, JSM-7500F, JEOL, Japan).

To observe the oxidation response properties of the hydrogels, 400 μL of CF gel and Ber@CF gel were immersed in 100 μL of deionized water and 0.5 % hydrogen peroxide solution, respectively, and photographed to observe the hydrogel state, which was continued to be photographed and observed after 12 h. The excess hydrogels were used for subsequent rheological tests.

To measure the rheology properties of hydrogels, the stress-controlled rheometer (HR-1, TA Instrument, USA) with a 20 mm flat plate was used. The storage modulus (G') and loss modulus (G'') were recorded at the strain of 1 %. The self-healing property of Ber@CF gel was measured by two cycles of step strain from 1 % to 250 %.

The hydrogel was extruded from a 1 mL syringe into a weighing vessel to observe the injectability of the hydrogel.

2.6. Drug release profile of Ber@CF gel

To study the release kinetics of Ber@CF gel, 200 μL of Ber@CF gel was immersed in 2 mL of aqueous solution and 0.5 % hydrogen peroxide solution, respectively, at 37 °C, and 20 μL of supernatant was collected at predetermined time intervals. In order to detect the amount of released berberine, 180 μL of DMSO was added to the extraction solution, and then the absorbance of the solution at 350 nm was read using an enzyme meter, and the cumulative release rate of berberine at different time points was calculated.

2.7. Antioxidative efficiency of Ber@CF gel

The antioxidative efficiency of Ber@CF gel were evaluated by DPPH free radical scavenging assay [48]. Firstly, DPPH was dissolved in the ethanol to the concentration of 100 μM . Subsequently, different concentration of Cur/HH gel dispersions were mixed with DPPH solution at the ratio of 1:1 (v/v), respectively. The mixture was oscillated and incubated in the dark at 37 °C for half an hour. Finally, the absorbance of reaction solution was measured at 516 nm. The free radical scavenging efficiency was calculated according to the following equation:

$$\text{DPPH scavenging (\%)} = ([A]_{\text{Blank}} - [A]_{\text{sample}}) / [A]_{\text{Blank}} \times 100\%$$

where $[A]_{\text{Blank}}$ was the absorption of the blank (DPPH + ethanol), and $[A]_{\text{sample}}$ was the absorption of the sample (DPPH + ethanol + sample).

2.8. Cell viability assay

L929 cells were cultured in RPMI-1640 medium supplement with 10 % FBS and 1 % P/S. 1×10^4 L929 cells per well were seeded into the 96-well plate with 100 μL of cell medium containing 10 % FBS and cultured for 24 h under a typical condition. The medium was replaced with a fresh medium containing predetermined concentrations of Ber@CF gel. Unless otherwise stated, the treatment of hydrogels in the cell experiments in this article is extraction. After incubated for another 24 h, MTT assay was used to investigate cell survival. The optical density at 570 nm was measured using the plate reader. The cell viability (%) was calculated according to the following equation:

$$\text{Cell viability (\%)} = ([A]_{\text{test}} - [A]_{\text{neg}}) / ([A]_{\text{pos}} - [A]_{\text{neg}}) \times 100\%$$

where $[A]_{\text{Control}}$ was the absorption of the control wells without hydrogels, and $[A]_{\text{sample}}$ was the absorption of the wells with hydrogels.

L929 cells were inoculated in 12-well plates at a cell density of 1×10^5 cells/well, and after 24 h of incubation, the blank group was replaced with new medium to continue the incubation, one group was added with Ber micelles, and the rest of the groups were intervened against CF gel and Ber@CF gel extracts (both containing 20 $\mu\text{g/mL}$ of

berberine), respectively. After 12 h of co-incubation, the cells were stained using a live/dead staining kit and observed with a Thunder (Leica, Germany) microscope. The fluorescent areas of the photographs were quantitatively analyzed using the software Image J and counted according to the following formula:

$$\text{Cell viability (\%)} = [A]_{\text{live}} / ([A]_{\text{live}} + [A]_{\text{dead}}) \times 100\%$$

where $[A]_{\text{live}}$ is the area of the region representing live cells in green and $[A]_{\text{dead}}$ is the area of the region representing dead cells in red.

In the cellular experiments of glucose damage reduction by hydrogel, glucose at a final concentration of 50 mM was added to the culture medium of three groups except the Blank group, and Ber-F127 micellar solution and the extract of Ber@CF gel were added to two of them as experimental groups. The subsequent steps of live/dead staining steps and results analysis were as above.

2.9. In vitro antioxidant properties

L929 cells were inoculated into 12-well plates at a cell density of 1×10^5 cells/well, and after 24 h of incubation, the blank group was replaced with fresh medium to continue incubation, and the remaining groups replaced the original medium with 1 mL of fresh medium containing 100 μM H_2O_2 to simulate the environment of high oxidative stress, while two groups of medium were supplemented with Ber micelles and Ber@CF gel immersion, respectively (both containing 20 $\mu\text{g/mL}$ berberine). After 12 h of co-incubation, cells were stained with a live/dead staining kit and observed with a Thunder microscope. The fluorescent areas of the photographs were quantitatively analyzed using the software Image J and counted according to the following formula:

$$\text{Cell viability (\%)} = [A]_{\text{live}} / ([A]_{\text{live}} + [A]_{\text{dead}}) \times 100\%$$

where $[A]_{\text{live}}$ is the area of the region representing live cells in green and $[A]_{\text{dead}}$ is the area of the region representing dead cells in red.

2.10. In vitro anti-high-glucose damage assay

L929 cells were inoculated into 12-well plates at a cell density of 1×10^5 cells/well, and after 24 h of incubation, the blank group was replaced with fresh medium to continue incubation, and the remaining groups replaced the original medium with 1 mL of fresh medium containing 50 mM glucose to simulate the environment of high glucose damage, while two groups of medium were supplemented with Ber micelles and Ber@CF gel immersion, respectively (both containing 20 $\mu\text{g/mL}$ berberine). After 12 h of co-incubation, cells were stained with a live/dead staining kit and observed with a Thunder microscope. The fluorescent areas of the photographs were quantitatively analyzed using the software Image J and counted according to the following formula:

$$\text{Cell viability (\%)} = [A]_{\text{live}} / ([A]_{\text{live}} + [A]_{\text{dead}}) \times 100\%$$

where $[A]_{\text{live}}$ is the area of the region representing live cells in green and $[A]_{\text{dead}}$ is the area of the region representing dead cells in red.

2.11. Exploration of in vitro anti-inflammatory properties

The cells used in the in vitro anti-inflammatory experiments were a mouse monocyte-macrophage cell line (RAW 264.7) cultured in a constant temperature cell culture incubator at 37 °C with continuous 5 % carbon dioxide ventilation using DMEM high glucose medium.

To mimic a highly inflammatory environment, macrophages were induced with lipopolysaccharide (LPS). RAW 264.7 cells were inoculated into 24-well plates at a cell density of 1×10^5 cells/well and, after 24 h of incubation, the cells were induced by replacing the original medium with 500 μL of fresh medium containing 500 ng/mL LPS. Meanwhile, two groups of cells were additionally treated with Ber-F127 and Ber@CF gel (both containing 10 $\mu\text{g/mL}$ berberine). Normal medium

was used as a blank control. After 24 h of co-incubation, the supernatant was discarded, washed twice with PBS, and cells were labeled for observation using the ROS probe 2',7'-dichlorofluorescein diacetate (DCFH-DA). The cell culture procedure was the same as described above, and after 24 h of co-incubation of the cells with the LPS and materials, the supernatant was discarded and washed twice with PBS, and 100 μ L of the DCFH-DA concentration of 100 μ M was added to each well. DCFH-DA solution at a concentration of 100 μ M was added to each well, and the cells were incubated at 37 °C for 30 min in the dark. The wells were then washed twice with PBS, photographed and observed with a Thunder microscope.

The effect of Ber@CF gel on macrophage phenotype in inflammatory environment was detected by flow cytometry. RAW 264.7 The cell culture procedure was the same as described above, and the cells were collected and centrifuged after 24 h of co-incubation with LPS and the materials, resuspended and washed with PBS solution and centrifuged again. The cells were then incubated at 4 °C under light protection with PE-labeled CD206 antibody. In addition, the supernatant was collected for the determination of the levels of the inflammatory factor TNF- α and the anti-inflammatory factor IL-10 by using the Elisa kits.

Ber@CF gel reduction of AGE production was tested with L929 cells. Specific cells were cultured and grouped as in section 2.9. Cells obtained from the co-culture were lysed, centrifuged and protein suspension was collected and AGE levels were determined using an Elisa kit.

The inhibitory effect of Ber@CF gel on the EIF2AK2 pathway was obtained by immunofluorescence staining. Similarly LPS was used to stimulate RAW264.7 macrophages as a control group, and a blank group and a corresponding material group were set up. RAW264.7 macrophages obtained by co-culturing for 24 h were fixed with 4 % paraformaldehyde for 30 min, after which macrophages were treated with 0.5 % Triton-X100 for 20 min and closed with 10 % goat serum for 30 min. Subsequently, the cells were incubated with anti-EIF2AK2 antibody at 4 °C for one night, followed by the addition of Cy3 conjugated Goat Anti-Rabbit IgG H&L to bind the primary antibody at 37 °C for 1 h. The cells were then stained with DAPI staining for 5 min, and then photographed with a Thunder microscope to observe the fluorescence intensity.

2.12. Full-thickness wound healing study in diabetic mice

The male C57BL/6J mice were induced to diabetes by STZ. Briefly, after fasting for 12 h, the mice were received intraperitoneal injections with STZ in citrate buffer solution (0.1 M, pH 4.5) at a dose of 160 mg/kg body weight. The blood glucose and weight of each mouse were measured, and all mice exhibited severe hyperglycemia (>16.7 mM) for a week. 36 mice were randomly divided into four groups (9 mice per group): normal mice (Non-DM), diabetic mice (DM Control), hydrogel group (CF gel) and drug-loaded hydrogel group (Ber@CF gel). The mice were anesthetized with isoflurane and the hair on their backs was shaved. Two 8 mm circular full-thickness wounds were made on each side of the back and then 50 μ L prepared materials were respectively applied to the wounds after the surgery. Ultimately, the wound surface was covered with the 3 M Tegaderm TM film. On Day 0,2,4,7,10 and 14, the wounds were captured by a digital camera to evaluate the process of wound healing and the dressings were refreshed. The wound healing ratio was calculated according to the following equation:

$$\text{Wound healing ratio (\%)} = ([A]_0 - [A]_t)/[A]_0 \times 100\%$$

where [A]₀ was the initial wound area on Day 0 and [A]_t was the wound area on the indicated day.

In addition, to monitoring ROS scavenging at the wound site, *in vivo* fluorescence images were obtained using an *in vivo* imaging system (Caliper Life Sciences, USA) by DCFH-DA labeling on Day 0, 2 and 4.

2.13. Histological analyses

On Day 2, 4 and 14, 2 mice from each group were sacrificed and their tissue samples were harvested. Half of the tissues from days 2 and 4 were immersed in 4 % paraformaldehyde all for subsequent tissue staining, and half were treated with liquid nitrogen and stored at −80 °C for Elisa protein assays. On Day 14, all the mice were sacrificed and their tissue samples were harvested.

Tissues fixed with 4 % paraformaldehyde were embedded in paraffin, sectioned according to standard methods and stained. Among them, tissues on day 14 were stained with H&E staining, Masson staining and Sirius scarlet staining, tissues on day 2 were stained with immunofluorescence staining (F4/80 and CD206 expression), and tissues on day 4 were stained with immunofluorescence staining (TNF- α , IL-10, F4/80 and CD206 expression) and immunohistochemical staining (EIF2AK2 expression). Images of H&E staining, Masson staining and Sirius red staining were taken with a microscope (Leica DM6B, Germany). Wound epithelial gap, wound edge and epithelial thickness were measured and counted using ImageJ 1.52a. All immunofluorescence staining and immunohistochemistry images were recorded by Panoramic MIDI (3D HISTECH, Hungary), wound edges and wound epithelial thickness were measured and counted by ImageJ 1.52a, and immunofluorescence staining-positive cells were quantitatively analyzed by ImageJ.

Tissue homogenates stored at −80 °C were centrifuged and the supernatant was collected, and the corresponding protein levels of tissues on days 2 and 4 were measured using Elisa kits for TNF- α , IL-10, and AGE.

2.14. Statistical analysis

All data were obtained from at least three individual experiments and presented as mean $\pm$ standard deviation (mean $\pm$ SD). All experiments were analyzed for statistical significance using GraphPad Prism 7.0. Unpaired Student's t-test was used for two groups and one-way ANOVA followed by Bonferroni's post hoc test was used for more than two groups. The data were statistically significant when * p < 0.05. **** p $\leq$ 0.0001, *** p $\leq$ 0.001, ** p $\leq$ 0.01, * p $\leq$ 0.05.

3. Results and discussion

3.1. Preparation and characterization of Ber@CF gel

The low solubility of berberine represents a significant obstacle to its bioavailability. In order to enhance the solubility of berberine, FDA-approved F127 was employed to prepare micelles by wrapping berberine with F127 through the film hydration method. As illustrated in Fig. 1a, the free berberine solution exhibited a distinct precipitation phenomenon under standard conditions. In contrast, the Ber-F127 micelle solution was observed to be clear and transparent, with a yellow hue. The distribution of micelles in the solution was observed using the fluorescence property of berberine. As illustrated in Fig. S1, the Ber-F127 micellar solution exhibited uniform green fluorescence, indicating that the micelles were homogeneously dispersed in water. The micelles were observed to have a uniform spherical morphology under transmission electron microscopy (TEM), with a particle size of 22.68 nm as determined by dynamic light scattering (DLS) (Fig. 1b). The absorption peak of berberine was determined to be 350 nm (Fig. S2), allowing the loading and encapsulation of berberine by Ber-F127 micelles to be calculated as 3.67 % and 77.00 %, respectively (Table S1). These results indicate that the Ber-F127 micelles have an appropriate particle size and an excellent loading efficiency.

It is report that berberine primarily targets the intracellular EIF2AK2 protein to inhibit inflammatory processes [49]. To demonstrate the higher bioavailability of Ber-F127 micelles, we employed a fluorescence labelling technique to observe the intracellular distribution of berberine.

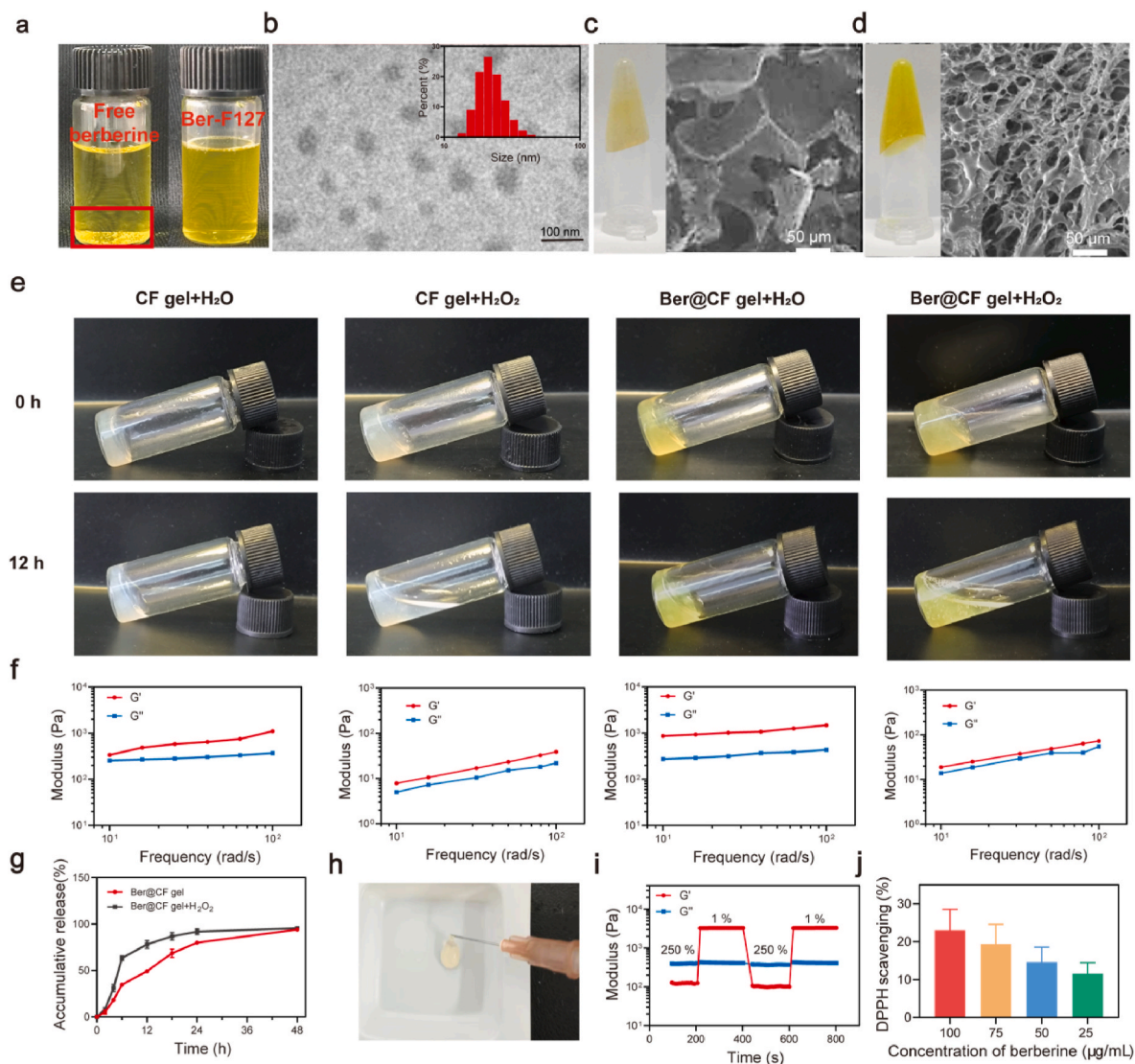


Fig. 1. Synthesis and Characterization of Ber-F127 micelles, CF gel and Ber@CF gel: (a) Comparison photo of berberine suspension (left) and Ber-F127 micelle solution (right), with the precipitation of berberine in the red box; (b) TEM photo of Ber-F127 micelles and the particle size distribution; (c–d) Macroscopic photos and microscopic SEM photographs of CF gel (c) and Ber@CF gel (d); (e) Photographs of the state of hydrogel at 0 h and 12 h in aqueous solution and 0.5 % H_2O_2 solution; (f) Rheological characterization of hydrogel at 12 h in aqueous and 0.5 % H_2O_2 solutions; (g) Drug release profile of Ber@CF gel in aqueous and hydrogen peroxide solutions; (h) Photographs of Ber@CF gel extrusion from syringe; (i) Cyclic strain rheology test of Ber@CF gel; (j) DPPH scavenging assay of Ber@CF gel.

As demonstrated in Fig. S3, the cell fluorescence photographs and the quantitative test of intracellular berberine fluorescence intensity revealed that the fluorescence intensity of berberine in Ber-F127 micelles-treated cells was significantly elevated, accompanied by evident cell phagocytosis. This suggests that Ber-F127 micelles facilitate the cellular uptake of free berberine. Furthermore, the expression of the EIF2AK2 protein within the cells was observed following the entry of the micelles. It was observed that the Ber-F127 micelles group exhibited a notable reduction in the level of EIF2AK2-related proteins in LPS-stimulated RAW264.7 macrophages (Fig. S4). Additionally, the *in vivo* experiments on diabetic wounds demonstrated that Ber-F127 micelles more effectively reduced the level of traumatic ROS in comparison to free berberine (Fig. S5), particularly on day 4, when a notable discrepancy was observed in ROS levels between the two groups. The preceding results indicate that Ber-F127 micelles exhibit enhanced bioavailability and a superior therapeutic efficacy compared to free berberine.

To further enhance the bioavailability of Ber-F127 micelles in highly inflamed wounds, an oxidation-responsive supramolecular hydrogel

was prepared for the purpose of loading Ber-F127 micelles. Hyaluronic acid backbones grafted with cyclodextrin molecules (HA-CD) and ferrocene molecules (HA-Fc) were prepared according to our previously reported procedure [47]. The successful synthesis of these materials was confirmed by ^1H NMR spectroscopy (D_2O) (Fig. S6). A CF gel was prepared by mixing HA-CD and HA-Fc at a molar ratio of 1:1 (Fig. 1c), while a Ber@CF gel was prepared by adding Ber-F127 micelles to the CF gel (Fig. 1d). Both the CF gels and the Ber@CF gels could be stored in inverted centrifuge tubes. Additionally, the Ber@CF gels exhibited a yellow hue due to the incorporation of Ber-F127 micelles. The scanning electron microscopy (SEM) photographs revealed that both the CF gels and the Ber@CF gels exhibited a three-dimensional mesh structure. However, the pore sizes of the Ber@CF gels were denser, which could be attributed to the hydrogen bonding between the F127 micelles and the hyaluronic acid. Furthermore, the green fluorescence of Ber-F127 micelles was observed in the fluorescence photographs, which demonstrated uniform dispersion in the Ber@CF gels (Fig. S7). This indicated that the Ber@CF gels had been successfully prepared.

Reactive oxygen species (ROS) at diabetic wounds can react with

ferrocene groups to generate ferrocenyl salts (Fc^+) which facilitate the hydrogel's disassembly, thereby enabling responsive drug release. The oxidative responsiveness of Ber@CF gel can be replicated in a high-inflammatory environment through the addition of the 0.5 % H_2O_2 to the gel. As illustrated in Fig. 1e, the deionized water-treated hydrogels exhibited a similar liquid level orientation to the pre-treatment samples, with all liquid levels angled relative to the horizontal. However, following the addition of hydrogen peroxide, both the CF gel and the Ber@CF gel demonstrated a horizontal orientation of their liquid levels. Meanwhile, SEM photographs revealed that the pore size of the hydrogel was significantly enlarged after H_2O_2 treatment (Fig. S8). The modulus of the hydrogels following the oxidation reaction was subsequently evaluated using a plate rheometer. The storage modulus (G') and loss modulus (G'') of the hydrogen peroxide-treated hydrogels exhibited a

notable decline in comparison to the water-treated group, indicative of their dissociation in an oxidative environment. The release of Ber-F127 micelles before and after dissociation of Ber@CF gels was quantified using a UV-visible spectrophotometer. As illustrated in Fig. 1g, approximately 65 % of berberine was released from Ber@CF gels in the H_2O_2 solution within the first 6 h, with a release rate that was almost twice that of the control. The aforementioned results demonstrate that the Ber@CF gel exhibits excellent oxidative responsiveness, which facilitates the enhanced therapeutic efficacy of Ber-F127 micelles in diabetic wounds. In addition, *in vivo* imaging photographs showed that the Ber@CF gel maintained well in the wound and continued to release the drug for 24 h compared to free berberine and Ber-F127 micelles, thus ensuring that berberine would remain in the wound and continue to exert its pharmacological effects (Fig. S9).

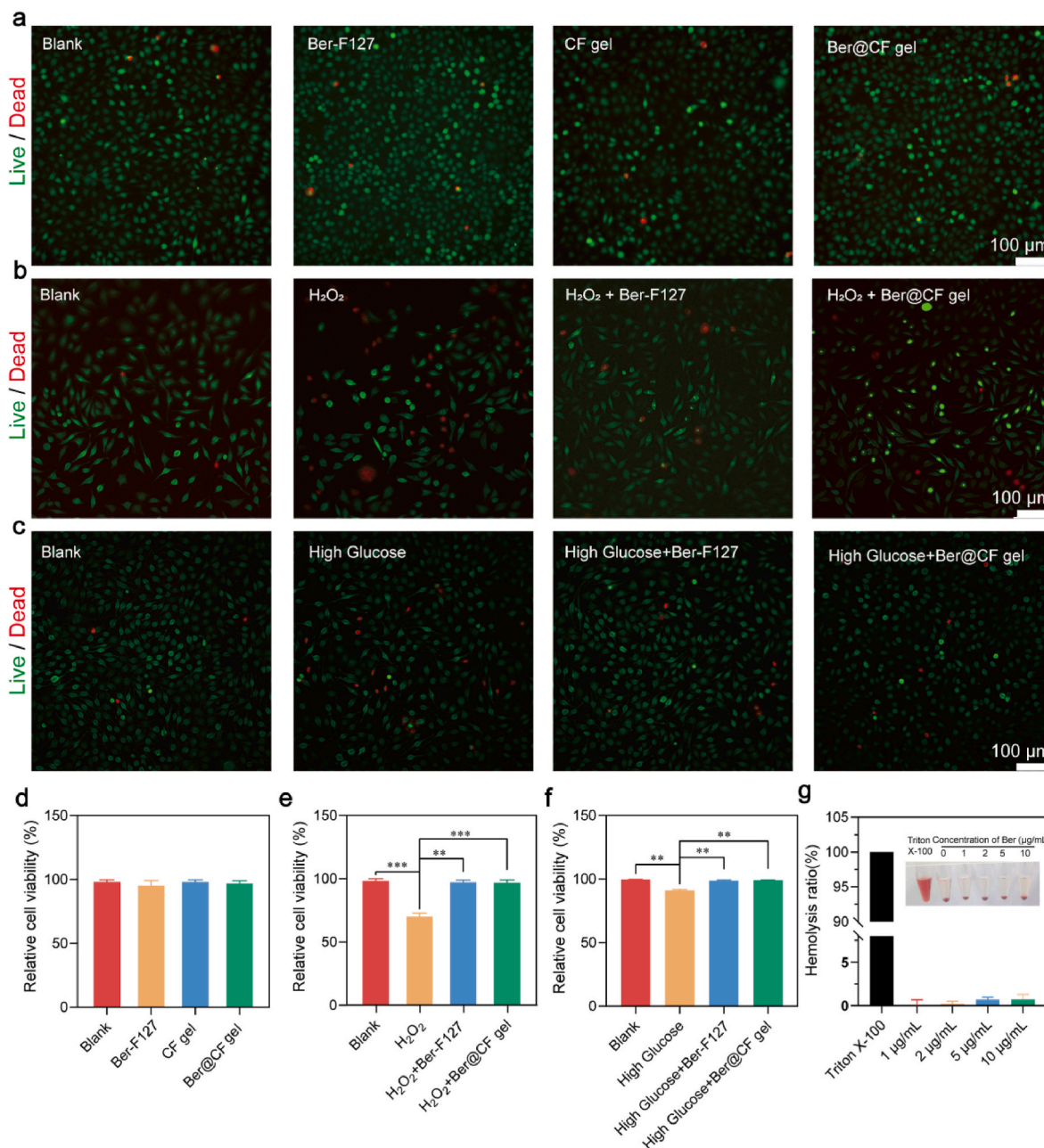


Fig. 2. *In vitro* biosafety tests of hydrogels: (a) Live/Dead staining photos and quantitative analysis(d) of L929 cells cultured with different materials for 24 h; (b) Live/Dead staining photos and quantitative analysis(e) of L929 cells cultured in 100 μM H_2O_2 environment with different materials for 24 h; (c) Live/Dead staining photos and quantitative analysis(f) of L929 cells cultured in 50 mM glucose environment with different materials for 24 h; (g) Hemolysis test photos and quantitative analysis.

The supramolecular structure of HA-Fc and HA-CD host-guest assembly also endows Ber@CF gel with favorable injectability and self-healing properties. As illustrated in Fig. 1h, Ber@CF gel could be readily extruded from a 1 ml syringe, thereby facilitating the filling of the wound with hydrogel. Meanwhile, the rheological assessment of the hydrogel revealed that it exhibited the capacity to revert to its gel state in the presence of minor deformations following extensive deformation, which resulted in its transformation into a sol (Fig. 1i). Furthermore, the methoxyphenol structure present in the berberine molecule endows it with the capacity to scavenge free radicals. Consequently, the Ber@CF gel is capable of scavenging DPPH free radicals (Fig. 1j), which enables the hydrogel to mitigate the pro-inflammatory environment characteristic of diabetic wounds.

3.2. *In vitro* biosafety assay of Ber@CF gel

It is imperative that hydrogel dressings are safe from a biosafety perspective before they can be used in subsequent stages of wound healing. High concentrations of berberine have been demonstrated to possess notable biotoxicity. Accordingly, the initial step involved co-culturing varying concentrations of Ber-F127 micelles with L929 cells, with the objective of identifying the optimal berberine concentration through the MTT method (Fig. S10). Subsequently, Ber-F127 micelles, CF gel, and Ber@CF gel were co-cultured with L929 cells and stained with Calcein -AM/Propidium Iodide (PI) to detect live cells (green) and dead cells (red). As illustrated in Fig. 2a, a substantial number of viable cells were observed in the live/dead staining images of each group. The quantitative analysis of fluorescence area demonstrated that the survival rate of cells in each group exceeded 95 % (Fig. 2d).

An excess of ROS in diabetic wounds has been demonstrated to cause cell death and inhibit the wound healing process. To simulate cell survival in diabetic wounds with a high concentration of ROS, H_2O_2 was added at a final concentration of 100 μM to the culture medium. The Ber-F127 micelles and Ber@CF gel were then co-cultured with L929 cells in an H_2O_2 solution, and the cells were stained for live/dead cells. As illustrated in Fig. 2b, the H_2O_2 group exhibited a considerable number of red dead cells, whereas the Ber-F127 micelles and Ber@CF gel group demonstrated a notable reduction in dead cells. Further quantitative analysis demonstrated that the cell survival rate declined to less than 70 % following the addition of H_2O_2 in comparison to the blank group. Conversely, the cell survival rate exceeded 95 % in both the Ber-F127 group and the Ber@CF gel group (Fig. 2e). Furthermore, damage to the cytoskeleton was observed through staining and labeling of the cytoskeleton and subsequent photography. As illustrated in Fig. S11, the addition of hydrogen peroxide resulted in the disappearance of the microfilament structure of the cytoskeleton, accompanied by a notable reduction in its number and fluorescence intensity. The microfilament structure of the cells exhibited a similar state after the addition of Ber-F127 micelles and Ber@CF gel as that observed in the blank group, indicating that the cells had been restored to a healthy state. The above results suggest that hydrogels can effectively mitigate apoptosis and cell damage induced by ROS.

The hyperglycemic microenvironment of diabetic wounds has been demonstrated to cause cellular damage in addition to the characteristic inflammatory response. Following the addition of glucose at a final concentration of 50 mM to the medium of L929 cells and subsequent co-culture with each material for 24 h, cell survival was observed through live/dead staining. As illustrated in Fig. 2c, the data demonstrated that a considerable number of red dead cells were present in the high glucose environment. However, the incorporation of Ber-F127 micellar solution and Ber@CF gel extract proved to be an effective strategy for reducing cell death. The results were subjected to quantitative and statistical analysis (Fig. 2f). In the high-sugar environment, the cell survival rate decreased to approximately 80 %, whereas the Ber-F127 micellar solution and Ber@CF gel increased the cell survival rate to over 90 %. These findings indicate that berberine-loaded hydrogels are effective in

reducing cell apoptosis in high-sugar environments.

Hemocompatibility is also one of the most important indicators of biosafety of dressings in clinical practice. To demonstrate that the Ber@CF gel has good hemocompatibility, a hemolysis test was conducted on hydrogel extracts in PBS buffer. As illustrated in Fig. 2g, Triton X-100 disrupts the cell membrane of erythrocytes, resulting in a reddish coloration of the supernatant. In contrast, the supernatant of the Ber@CF gel, containing varying concentrations of berberine, exhibited a colorless and transparent appearance. A further quantitative analysis, conducted by reading the absorbance of the supernatant at 545 nm, demonstrated that the hemolysis rate of the Ber@CF gel group was less than 5 %. This indicates that the Ber@CF gel has good blood compatibility and meets the requisite biosafety standards for use.

3.3. *In vitro* anti-oxidative and anti-inflammation assay of Ber@CF gel

In order to mitigate high inflammation in wounds, except avoiding apoptosis and injury in a highly oxidizing and glycolytic environment, reducing cellular production of ROS is also a proven means of alleviating oxidative stress. Lipopolysaccharide (LPS) is commonly used *in vitro* to mimic highly inflammatory physiological environments because it causes oxidative stress in cells, generates large amounts of ROS, induces highly inflammatory physiological environments, and inhibits the transformation of M2-type macrophages. RAW264.7 cells were cultured in LPS-containing medium to mimic the physiological environment of endogenous high inflammation and co-cultured with Ber-F127 and Ber@CF gel for 12 h. As shown in Fig. 3a, the ROS level in the LPS group was significantly higher than that in the blank group, whereas the fluorescence area in the Ber-F127 and Ber@CF gel groups was close to that in the blank group, indicating that the Ber@CF gel reduced intracellular ROS.

LPS-treated RAW264.7 macrophages were collected and labeled with an anti-CD206 antibody, and the resulting cells were analyzed using flow cytometry. It should be noted that the concentration of Ber-F127 micelles refers to the effective concentration of berberine in the micelles. As illustrated in Fig. 3b, the data demonstrated that the CD206 expression in macrophages exhibited a notable increase with the elevation of berberine concentration, from 10.48 % to 48.13 %. The CD206 expression level of the Ber@CF gel containing 20 $\mu g/mL$ berberine was also markedly elevated in comparison to that of the control group. This indicates that the Ber@CF gel can elevate the proportion of M2-type macrophages, thereby suppressing inflammation (Fig. 3f). The levels of inflammatory factors in the cell supernatant were further examined with an ELISA kit. Following LPS stimulation, the levels of TNF- α , a pro-inflammatory factor secreted by macrophages, were significantly increased (Fig. 3d–S12a). However, after the addition of Ber-F127 micelles and extracts of Ber@CF gel to the culture medium, the levels of TNF- α decreased to levels close to those of the blank group. In contrast to TNF- α , the level of IL-10, an anti-inflammatory factor secreted by the cells, exhibited a tendency to increase following treatment with Ber-F127 micelles and Ber@CF gel (Fig. 3e–S12b). This indicates that Ber@CF gel has the capacity to suppress inflammatory processes.

Furthermore, an immunolabeling assay was conducted to determine the level of EIF2AK2 protein, the target protein of berberine that inhibits inflammation, in the cells. As illustrated in Fig. 3c, the level of EIF2AK2 protein in macrophages increased significantly following LPS stimulation. Conversely, the level of EIF2AK2 protein decreased after treatment with Ber-F127 micelles and Ber@CF gel. The aforementioned results suggest that Ber@CF gel may regulate macrophages by influencing the EIF2AK2 protein, thereby achieving anti-inflammatory effects. Furthermore, a sustained hyperglycemic environment has been demonstrated to result in the generation of substantial quantities of advanced glycation end products (AGE), which have been shown to intensify inflammatory responses. Berberine has the ability to reduce AGE formation in high glucose environments thereby reducing

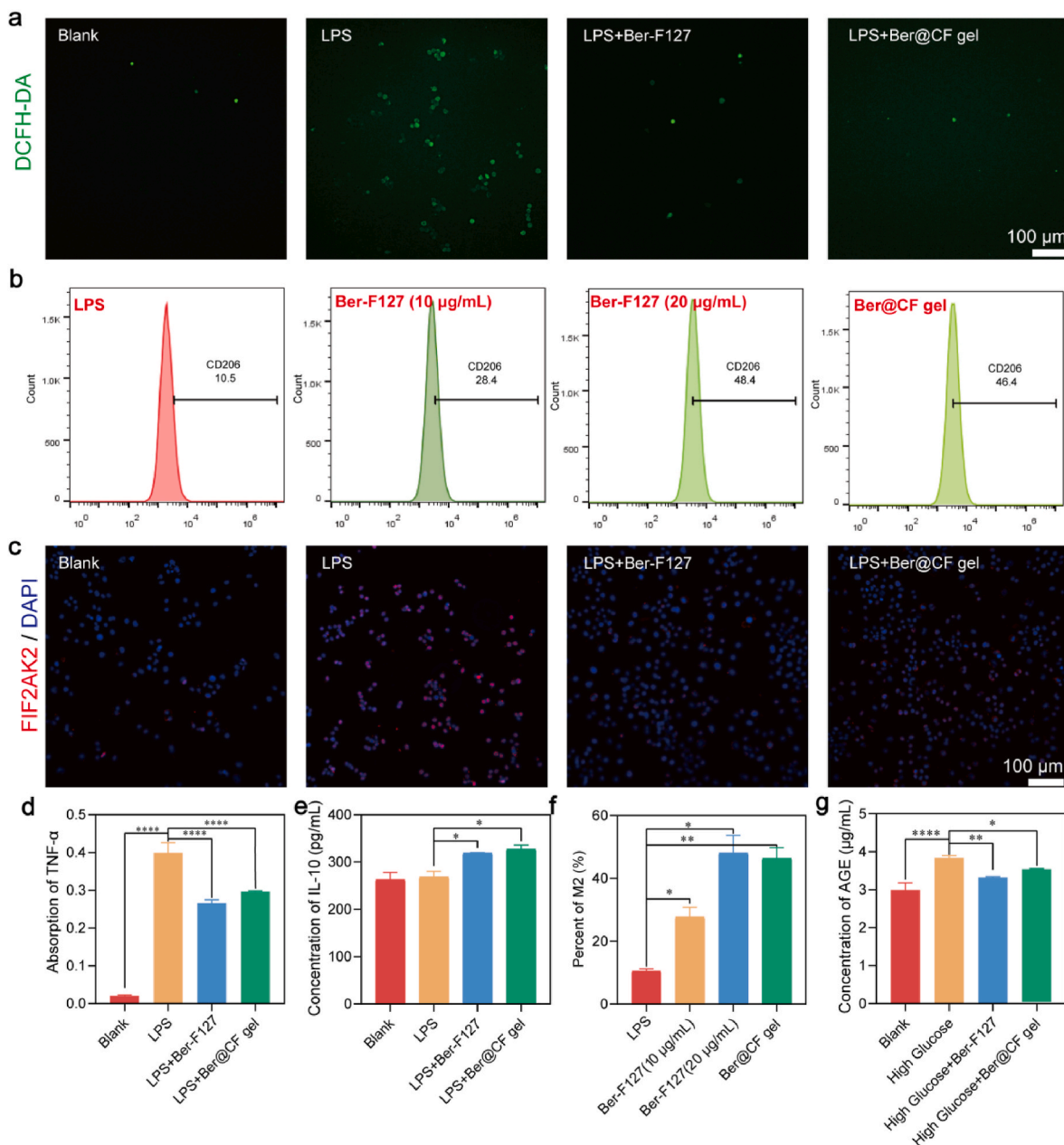


Fig. 3. *In vitro* simulation of anti-inflammation tests of hydrogels: (a) Intracellular ROS photos of DCFH-DA-labeled RAW264.7; (b) Flow cytometry results of CD206 in RAW264.7 cells and quantitative analysis (f) after co-cultivation with different materials for 24 h; (c) Immunofluorescence staining of EIF2AK2-related proteins in RAW264.7 macrophages after treatment with different materials for 24 h; (d) ELISA test results of TNF-α and IL-10 (e) levels secreted by RAW264.7 cells after LPS stimulation and co-cultivation with different materials for 24 h; (g) ELISA test results of intracellular AGE concentration after co-cultivation with different materials in high glucose environment for 24 h.

inflammation and decreasing associated damage [16]. The intracellular AGE content in the high glucose environment was quantified using an ELISA kit. As illustrated in Fig. 3G, the intracellular AGE content in the high glucose environment increased by approximately 3 µg/mL, whereas the addition of Ber-F127 micelles and hydrogels resulted in a reduction of the AGE content by approximately 1 µg/mL (Figure S12 c). These findings suggest that Ber@CF gel can modulate inflammation in multiple ways.

3.4. *In vivo* wound healing assays of Ber@CF gel

To demonstrate that Ber@CF gel can have the same excellent anti-inflammatory effect *in vivo*, we constructed a diabetic mouse model of

full-layer dorsal skin defects for further validation. The mice were randomly divided into four groups: a normal control group (non-DM), a diabetic control group (DM control), a pure hydrogel group (CF gel), and a drug hydrogel group (Ber@CF gel). The latter three groups were injected intraperitoneally with STZ to induce type I diabetes. The modeling was deemed successful if the mice exhibited a blood glucose level exceeding 16.7 mmol/L and a notable weight loss within a one-week period (Table S2). Subsequently, two symmetrical wounds of 8 mm in diameter were created on the backs of the mice, and the drugs were replaced and photographed at designated intervals (Fig. 4a). Meanwhile, the hydrogel was ensured to fill the wounds well due to its favorable injectability and capacity to conform to the shape of the wound (Fig. S13). The wounds were then uniformly wrapped with a

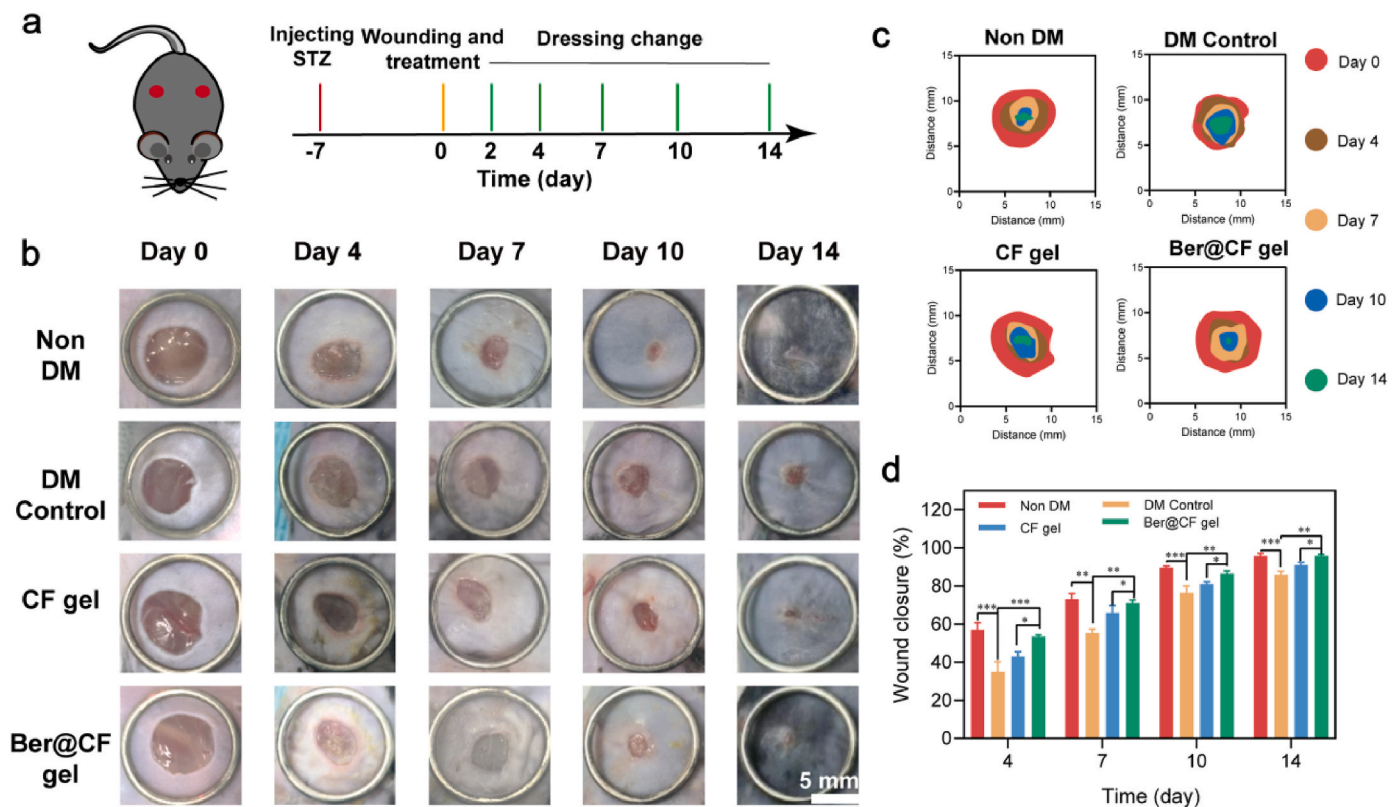


Fig. 4. Hydrogel-promoted chronic wound healing status tested in a mouse model of total skin defects on the back: (a) Schematic diagram of the process in the mouse model of total skin defects on the back; (b) Photographs of typical back wound healing in mice at different days; (c) Traces of wound contraction in mice at different days; (d) Statistics of wound healing rate in mice.

Tegaderm 3 M dressing patch on the outermost layer to prevent the hydrogel from dislodging and to prevent the introduction of foreign bodies into the wounds.

The animal wounds were photographed and analyzed for signs of wound contraction on days 4, 7, 10, and 14, respectively. As illustrated in Fig. 4b and c, the DM control group exhibited the poorest wound healing outcomes, followed by the CF gel group. In contrast, the non-DM group and the Ber@CF gel group demonstrated the most favorable outcomes, with wounds essentially healed and beginning to re-epithelialize by day 14. The rate of wound healing was quantified on different days (Fig. 4d). At the conclusion of the inflammatory phase, which occurred on day 4, the wound healing rate was less than 35 % in the non-DM group. In contrast, the wound healing rate exceeded 50 % in both the non-DM and Ber@CF gel groups. After day 7, the non-DM group exhibited accelerated wound healing, reaching approximately 86 % by day 14. Concurrently, the non-DM and Ber@CF gel groups demonstrated wound healing rates exceeding 95 %, while the CF gel group exhibited a rate exceeding 90 %. These findings suggest that CF gel may facilitate the healing of chronic wounds, although the effect is not pronounced. In contrast, Ber@CF gel has been observed to effectively restore the wound healing rate to a level comparable to that observed in normal mice. This indicates that the primary mechanism of action of Ber@CF gel in promoting wound healing is berberine.

3.5. In vivo anti-inflammation assays of Ber@CF gel

Reactive oxygen species (ROS) serve as a crucial indicator of the extent of inflammation during the wound healing process, effectively delineating the status of the inflammatory phase. Constructing oxidative-responsive hydrogels to modulate the rate of drug release by the level of ROS at the wound can efficiently utilize the biological functions of berberine and in turn downregulate ROS levels and reduce

inflammation [47]. On days 0, 2, and 4, ROS were fluorescently labeled with DCFH-DA in the skin wounds of the animals. The mean fluorescence intensity (MFI) of the wounds in living mice was then measured and quantitatively analyzed (Fig. 5a and b). On day 2, the ROS levels in the DM control and CF gel groups were higher than those in the non-DM group, indicating that diabetic mice exhibited heightened inflammatory responses in their wounds due to hyperglycemia compared to normal mice. In contrast, the ROS levels were markedly diminished in the Ber@CF gel group. On the fourth day, the level of inflammation in the DM control group remained elevated, indicating that diabetic mice were unable to regulate wound inflammation effectively. The ROS levels in the non-DM group, the CF gel group, and the Ber@CF gel group were significantly lower. Furthermore, in comparison to the second day, the ROS levels in the latter three groups demonstrated a notable decline. Notably, the ROS levels in the CF gel group exhibited a reduction of approximately 50 %. This is attributed to the fact that after the fourth day, the inflammatory phase of the wound is in the declining stage, which coincides with the initial stages of the resolution phase.

In order to analyze the mechanism by which hydrogel modulates inflammation and promotes wound healing at the wound site, wound tissues from mice on days 2 and 4 were immunofluorescently stained with antibodies labeled with F4/80 and CD206. The immunofluorescent markers F4/80 and CD206 are used to identify all macrophages and M2-type macrophages, respectively. The latter are known to inhibit inflammation and account for a significant proportion of macrophages. The level of wound inflammation can therefore be reflected by the intensity of F4/80 and CD206 fluorescence. As illustrated in Fig. 5c and d, the F4/80 fluorescence intensity of all groups was markedly elevated on day 2, indicating that all groups were in the acute inflammatory phase at this time. The CD206 fluorescence intensity of the Ber@CF gel group and the non-DM group was significantly higher than that of the other groups, suggesting that the inflammatory environment of the two groups was

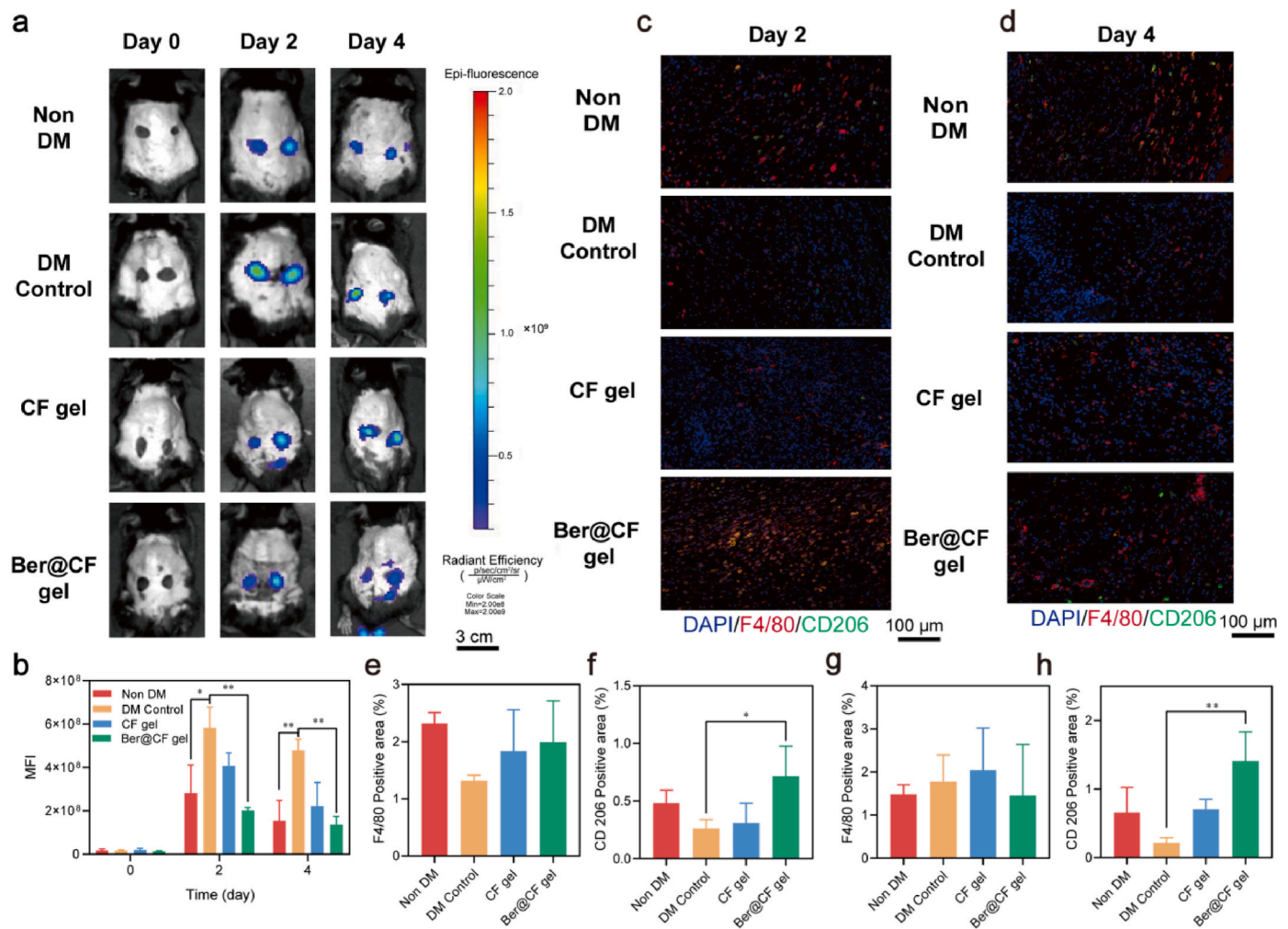


Fig. 5. Characterization of the anti-inflammatory effect of Ber@CF gel *in vivo*: (a) Representative fluorescence images of wounds from day 0 to day 4; (b) Statistical analysis of MFI values; Typical immunofluorescence photographs of F4/80 and CD206 on day 2 (c) and day 4 (d). Statistical results of F4/80-positive areas (e) and CD206-positive areas (f) on day 2; Statistical results of F4/80-positive areas (g) and CD206-positive areas (h) on day 4.

conductive to wound healing at this time. On day 4, the F4/80 fluorescence intensity of the DM control group and the CF gel group remained high, while the F4/80 fluorescence intensity of the Ber@CF gel group and the non-DM group exhibited a decline compared with that of day 2. The results were subjected to further quantitative and statistical analysis, and the findings are presented in Fig. 5e-h. On days 2 and 4, the F4/80 levels were elevated in all groups. However, the positive area of CD206 fluorescence intensity of the Ber@CF gel group was 0.72 % and 1.40 %, respectively, which was significantly higher than that of the DM control group (0.26 % and 0.21 %, respectively). These findings suggest that the Ber@CF gel can achieve inflammatory modulation by up-regulating the proportion of M2 macrophages.

Furthermore, immunofluorescence staining of the pro-inflammatory factor TNF- α and the anti-inflammatory factor IL-10, as well as immunohistochemical staining of the EIF2AK2 protein, were conducted on wound tissues on day 4, and the area of protein-positive tissue was quantified. As illustrated in Fig. 6a-f, the proportion of IL-10-positive area was markedly higher in the Ber@CF gel group and the non-DM group, at 21.75 % and 26.05 %, respectively. This was significantly greater than that observed in the remaining two groups. Conversely, the proportion of TNF- α -positive area was significantly higher in the DM control group and the CF gel group than in the remaining groups, at 7.53 % and 7.02 %, respectively. The immunohistochemical results demonstrated a notable decline in EIF2AK2 protein content in the Ber@CF gel group. The aforementioned results provide preliminary

evidence that the levels of inflammatory factors and related proteins in the tissues have decreased. To analyze the changes in factor concentrations at the wound site on different days after hydrogel treatment, the concentrations of IL-10, TNF- α , and AGE in the tissues on days 2 and 4 were quantified further by ELISA kit. The results are presented in Fig. 6g-k and Fig. S14. The highest concentration of TNF- α at both day 2 and day 4 was observed in the DM control group, at approximately 150 ng/mL, followed by the CF gel group. The lowest concentration was observed in the Ber@CF gel group and the non-DM group, both below 100 ng/mL. Conversely, the trend of the concentration of IL-10 was the opposite of TNF- α . Furthermore, the level of AGE was observed to decline in conjunction with a reduction in TNF- α concentration and an increase in IL-10 concentration. As demonstrated in Fig. S15, the immunohistochemistry results of RAGE on day 4 further substantiated that Ber@CF gel could down-regulate the level of RAGE, thereby indicating that hydrogels possess the capacity to curtail inflammation by removing AGE and down-regulating the level of RAGE protein. The aforementioned results indicated that the Ber@CF gel could effectively decrease the levels of AGE and EIF2AK2 protein, and further regulate the levels of tissue inflammatory factors, thereby achieving anti-inflammatory effects and promoting wound healing.

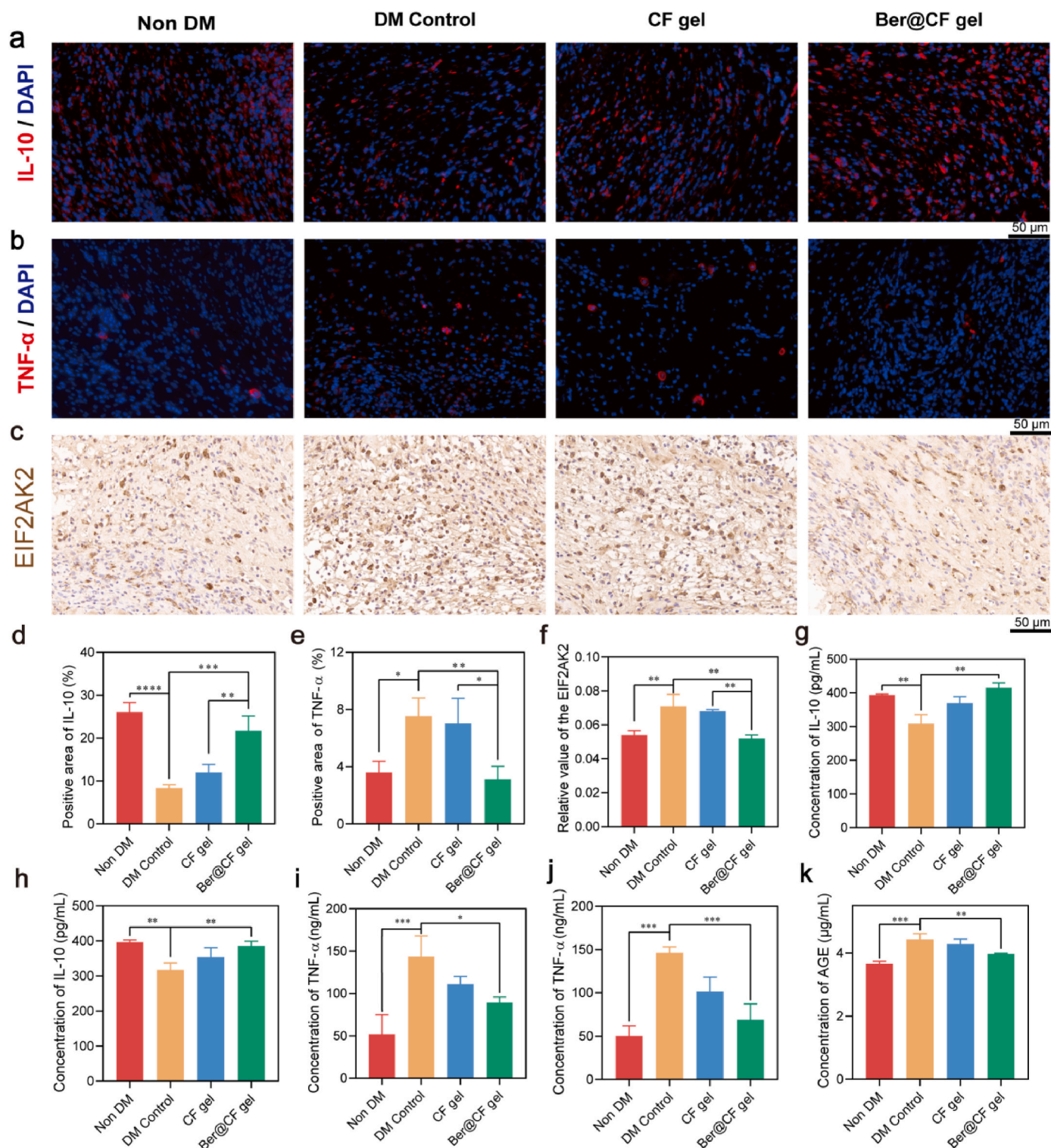


Fig. 6. Immunofluorescence photographs (a) of IL-10 and quantitative statistical results of positive areas (d); Immunofluorescence photographs (b) of TNF-α and quantitative statistical results of positive areas (e); Immunohistochemical photographs (c) of EIF2AK2 related proteins and quantitative statistics of positive areas (f); Elisa test concentration results for IL-10 on day 2 (g) and day 4 (h); Elisa test concentration results for TNF-α on day 2 (i) and day 4 (j); Elisa test concentration results for AGE on day 4 (k).

3.6. In vivo experiments with Ber@CF gel to improve the quality of tissue healing

Macroscopic photographic documentation of wounds is an inadequate representation of the true condition of the wound. To gain a more precise understanding of the impact of Ber@CF gel on the quality of

wound healing, we collected wound tissues from animals on day 14 and performed the corresponding histological staining. First, the results of the hemoglobin and eosin (H&E) staining demonstrated that the newborn granulation tissue remained defective (Fig. 7a). Consequently, further analysis was conducted on the wound healing and epidermal thickness in relation to the H&E staining results. The results are

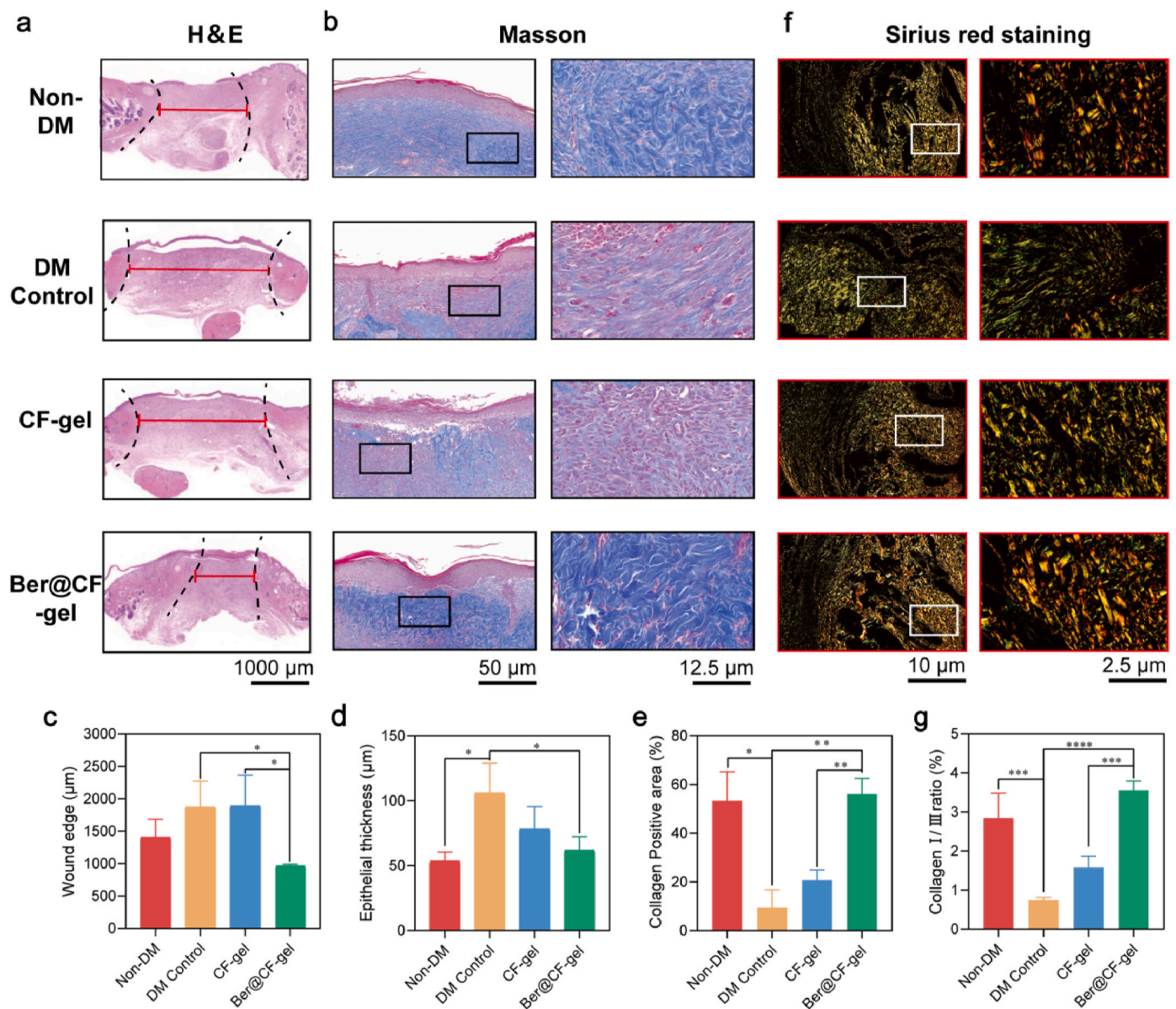


Fig. 7. Histological analysis of the effect of Ber@CF gel on wounds in diabetic mice on day 14: (a) Representative images of H&E staining of wound tissue; Quantitative analysis of epithelial gap (c) and epithelial thickness (d) from histological samples of H&E staining; (b) Representative images of Masson's staining of wound tissue; (e) Quantitative statistics of collagen positive area; (f) Representative images of Sirius red staining of wound tissue; (g) Quantitative statistics on the proportion of collagen type I-positive area and collagen type III-positive area.

presented in Fig. 7c and d. The width of the tissue wounds in the Ber@CF gel group was the narrowest, at approximately 1000 μm, which was similar to that of the non-DM group. Both of these groups exhibited a significantly improved width compared to the DM control group and the CF gel group, which was approximately 1500 μm. During the wound healing process, the epidermal thickness of the neoplastic tissue initially increases and then gradually decreases as the neoplastic granulation tissue optimizes. The epidermal thickness of the tissue wounds in the Ber@CF gel group was comparable to that of the tissue wounds in the non DM group, which was approximately 50 μm. Both were significantly thinner than that of the DM control group, followed by CF gel. This suggests that while the hyaluronic acid hydrogel was unable to resolve the damage caused by inflammation to the tissues, it was nevertheless capable of effectively enhancing the quality of wound healing.

Subsequently, Masson's trichrome staining was performed on the tissues to analyze the status of collagen deposition. As the wound healing process continues, the extracellular matrix undergoes

remodeling, accompanied by collagen deposition. This enables the new skin to regain its pre-trauma state. As illustrated in Fig. 7b, a considerable quantity of grossly distorted collagen was observed in the non-DM and Ber@CF gel groups, whereas no evidence of collagen was discernible in the DM control and CF gel groups. Further quantification of collagen-positive areas revealed that the proportion of collagen-positive areas in the Ber@CF gel group was approximately 50 %, whereas the proportion of collagen-positive areas in the DM control group and the CF gel group was less than 20 %. This indicates that the Ber@CF gel is capable of effectively promoting collagen deposition.

Collagen is comprised of multiple types that serve distinct functions at varying stages of wound healing. In general, as granulation tissue is optimized during the wound healing process, the amount of type I collagen gradually increases and replaces type III collagen. This results in an increase in the strength of the wound tissue, thereby improving the quality of wound healing. The tissue collagen types were labeled and quantified through the use of Sirius red staining. As illustrated in Fig. 7f

and g, the DM control group exhibited the highest level of type III collagen, while the other groups demonstrated elevated levels of type I collagen, with approximately four times the amount of type I collagen relative to type III collagen observed in the non-DM group and the Ber@CF gel group. Conversely, the DM control group displayed a lower ratio of type I to type III collagen, indicative of the poorest collagen status. Subsequently, the CF gel group exhibited a slight predominance of type I collagen over type III. The aforementioned data indicate that Ber@CF gel can effectively promote the maturation of wound tissue and enhance the quality of wound healing.

4. Conclusion

In this study, we successfully developed a Ber@CF hydrogel dressing with good anti-inflammatory properties, excellent promotion of chronic wound healing, and improved healing quality. The Ber-F127 micelles prepared by the hydrated membrane method improved the water solubility of berberine and promoted cellular cytosol, and the micellized berberine was shown to have better inhibition of inflammation in cell and animal experiments. The Ber@CF hydrogel has good oxidative reactivity, which promotes the responsive release of Ber-F127 micelles in diabetic wounds with high oxidative stress, and in addition, the hydrogel is injectable and fills the shape of the wound. In an in vitro cell simulation assay, the Ber@CF hydrogel showed good biocompatibility and enhanced cell survival in a high oxidative and high glyceic environment, reduced intracellular ROS production through the EIF2AK2 pathway, promoted macrophage differentiation to M2 type, reduced TNF- α and AGE levels, and increased IL-10 levels. Using a mouse model of full-thickness skin defects, it was further demonstrated that Ber@CF hydrogel promoted wound healing, improved the quality of wound healing, inhibited the production of ROS, promoted the transformation of macrophages to inhibit inflammation, and reduced the production of inflammatory factors. The hydrogel dressing developed in this study provides a reference for the design of clinical dressings that promote chronic wound healing by modulating inflammation.

CRediT authorship contribution statement

Xiaoyang Liang: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Honggui Chen:** Formal analysis, Data curation. **Rui Zhang:** Methodology, Investigation. **Zhixuan Xu:** Methodology, Investigation. **Guo Zhang:** Methodology, Investigation. **Chen Xu:** Writing – review & editing, Validation, Project administration. **Yang Li:** Writing – review & editing, Project administration, Funding acquisition, Data curation, Conceptualization. **Lei Zhang:** Writing – review & editing, Conceptualization. **Fu-Jian Xu:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.biomaterials.2024.123076>.

Data availability

Data will be made available on request.

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